

Research article

Computational epitope profiling and AI-driven protein engineering enable rational design of multi-epitope vaccines against *Mycobacterium tuberculosis*Xinfeng Li^a, Xinyu Tao^a, Mingyue Zhong^a, Yiyao Wang^a, Heng Xue^a, Binda T. Andongma^b, Shan-Ho Chou^b, Hongping Wei^a, Jin He^{b,*}, Hang Yang^{a,c,**}^a State Key Laboratory of Virology and Biosafety, Wuhan Institute of Virology, Chinese Academy of Sciences, Wuhan 430071, China^b National Key Laboratory of Agricultural Microbiology & Hubei Hongshan Laboratory, College of Life Science and Technology, Huazhong Agricultural University, Wuhan 430070, China^c Hubei Jiangxia Laboratory, Wuhan 430200, China

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ABSTRACT

Tuberculosis (TB), caused by *Mycobacterium tuberculosis* (Mtb), remains a major global health threat, accounting for approximately 1.5 million deaths annually. The rise of antibiotic-resistant strains further complicates treatment efforts. While vaccination is a cornerstone of disease control, the only licensed TB vaccine, Bacille Calmette-Guérin (BCG), shows limited efficacy in adults. There is thus a critical need for more effective vaccines. Multi-epitope vaccines, which incorporate key epitopes from multiple antigens, offer a promising strategy by eliciting both humoral and cellular immunity. Here, we employed a comparative epitopomics approach to identify immunodominant epitopes from eight major Mtb antigens and selected 17 potent epitopes for the design of a multi-epitope antigen. Using AI-driven protein design, we systematically optimized epitope arrangement and flanking sequences to generate a stable, structurally integrated antigen—MtbEpi-17. Computational analyses suggest that MtbEpi-17 can effectively interact with TLR2 and TLR4, potentially stimulating robust innate and adaptive immune responses. Our study provides a rational design framework for multi-epitope vaccines, and proposes MtbEpi-17 as a strong candidate for further preclinical and clinical evaluation.

1. Introduction

Tuberculosis (TB) is a zoonotic disease caused by *Mycobacterium tuberculosis* (Mtb) that kills millions of people worldwide each year. Current TB treatment involves a combination of multiple drugs over a 6- to 24-month course [1]. Although multidrug combination therapy has made progress in controlling TB, the emergence of multidrug-resistant (MDR) and extensively drug-resistant (XDR) strains poses an increasing challenge to these treatments [2]. Vaccination is an effective means of defense against pathogens [3]; however, the only licensed vaccine against TB, Bacillus Calmette-Guérin (BCG), demonstrates highly variable efficacy in adults (ranging from 0 % to 80 %) and offers limited protection, particularly in tropical and subtropical regions [4,5]. This variability has been attributed to several factors, including pre-existing immunity to environmental mycobacteria, insufficient diversity in T-cell responses due to human leukocyte antigen (HLA) heterogeneity, and the absence of key RD1-region antigens (e.g., ESAT-6

and CFP-10) in BCG that are critical for CD8⁺ T-cell activation [4,5]. Therefore, the development of novel, highly effective TB vaccines has become an urgent priority.

Multiple immunodominant antigens of Mtb, such as Ag85A, Ag85B, ESAT-6, CFP10, and HspX, are key targets for immune recognition [6–8]. As a result, vaccines that incorporate multiple antigens are promising candidates for achieving effective protection. However, developing such multi-immunogen vaccines presents significant challenges, including high production costs, technical complexity, and the difficulty of eliciting a balanced immune response across all component antigens [9]. Epitopes are the basic units of protein antigenicity and are the key structural and functional elements that trigger host-specific immune responses [10]. By integrating multiple epitopes into a single chimeric protein, it is possible to enhance the immune response and improve vaccine efficacy [11–14]. Currently, this technology has been widely applied in the design of various vaccines, including those targeting Mtb [15–23].

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The design of multi-epitope vaccines hinges on two critical factors: the selection of appropriate epitopes and their rational assembly into a single antigen. Current efforts in epitope selection rely heavily on bioinformatic predictions [22,24–28], with few studies incorporating experimentally validated epitopes, largely due to the convenience of computational methods. While advances in immunoinformatics have steadily improved the accuracy and ease of epitope prediction [29–31], these *in silico* forecasts may not always match the efficacy of empirically confirmed epitopes. To address this limitation, we recently developed a comparative epitomomics strategy that leverages the Immune Epitope Database (IEDB) [32], combining comparative proteomics with computational methods to rapidly identify characterized epitopes from specific antigens or pathogens. This approach provides a novel method for screening potent epitopes for vaccine design. Beyond epitope selection, a major unresolved challenge is the construction of the multi-epitope antigen. There are no established guidelines for optimally linking selected epitopes. The common practice of simply concatenating them in tandem often results in poor protein stability and solubility, which can ultimately compromise the immunogenicity of the vaccine. Therefore, developing a robust and rational strategy for epitope linkage is an urgent need in the field.

Here, we systematically analyzed the most prevalent Mtb antigens and focused on eight immunodominant candidates, including Ag85A (UniProt ID: P9WQP3), Ag85B (P9WQP1), ESAT-6 (P9WNK7), CFP10 (P9WNK5), Hsp65 (P9WPE7), Mpt64 (P9WIN9), HspX (P9WMK1), and Hsp70 (P9WMJ9). This study aims to identify potent epitopes from these antigens and leverage an artificial intelligence (AI)-driven protein engineering approach to design a structurally stable multi-epitope chimeric antigen. The innovative strategies are expected to provide critical insights into the development of novel TB vaccines and contribute to enhancing global TB control strategies.

2. Materials and method

2.1. Epitope screening based on comparative epitomomics

The reference protein sequences of the eight target antigens were obtained from the UniProt database (<https://www.uniprot.org/>) [33]. Experimentally validated human-recognized epitopes were obtained from IEDB (<https://www.iedb.org/>) [30]. To identify validated epitopes in these antigens, comparative epitomomics analysis was conducted using BLASTp [32]. In this analysis, the protein sequence of each antigen was used as the subject database, while the human recognized epitopes curated by IEDB were used as query sequences. Alignments were performed in BLASTp-short mode, optimized for short peptide sequences (typically ≤ 30 amino acids), with the following parameters: E-value cutoff of 1000 to facilitate identification of potentially relevant alignments for short peptides; word size of 2 to enhance sensitivity; BLOSUM62 scoring matrix (default for protein BLAST); gap opening penalty of 11; gap extension penalty of 1; and no composition-based statistics adjustment, as per the BLASTp-short default settings. Hits were filtered based on three criteria: (1) sequence identity $\geq 95\%$ to the query epitope, (2) query coverage $\geq 95\%$ of the epitope length, and (3) epitope length ≥ 6 amino acids.

For the retained epitopes, two key parameters were extracted from IEDB: “the number of references reporting the epitope” and “the number of positive assays including this epitope”. To quantify epitope effectiveness, the epitope enrichment index was introduced, defined as enrichment score = (number of references) $\times$ (number of positive assays). The raw enrichment scores were normalized to a 100-point scale by identifying the maximum raw enrichment score among all epitopes, setting this value to 100, and calculating normalized scores for other epitopes using the formula: normalized score = (epitope raw score / maximum raw score) $\times 100$. This normalization enables relative comparison of immunological support across epitopes, with a score of 100 representing the epitope with the most combined experimental

evidence. Epitope screening results were visualized using a bubble chart, with epitopes in the upper-right quadrant designated as potential high-efficacy epitopes.

The potential high-efficacy epitopes underwent a comprehensive manual evaluation to identify the most promising candidates, designated as potent epitopes, for multi-epitope vaccine construction. This assessment was based on objective and quantifiable immunological performance criteria derived from experimental assays available in IEDB. For each antigen, a selection of 0–2 epitopes was made, with one chosen as the norm; however, 2 epitopes were selected if they demonstrated comparably strong performance, while none were chosen in cases of weak performance. This approach aimed to ensure comprehensive antigen coverage while prioritizing immunologically robust candidates. Specifically, CTL epitopes were evaluated for several key parameters, including binding affinity to MHC class I molecules, specificity for MHC-I alleles, IFN- γ release activity, and cytotoxic potential. For HTL epitopes, the evaluation focused on their ability to stimulate important cytokines such as IFN- γ , IL-4, IL-2, TNF- α , and IL-13. Additionally, B-cell epitopes were scrutinized for their binding affinity to B-cell receptors (BCRs) as well as their protective efficacy in the context of pathogen challenges.

2.2. Rational design of multi-epitope antigens driven by artificial intelligence

The 17 epitope sequences were divided into three different functional modules: a CTL epitope subunit, an HTL epitope subunit, and a B-cell epitope subunit. Each subunit module served as an independent design template and was processed using the ProteinGenerator (PG) algorithm [34]. This approach enabled the computational generation of a series of candidate subunit structures. These structures were rigorously evaluated using per-residue local distance difference test (pLDDT) scores, a key indicator of structural prediction quality [35]. Specifically, pLDDT values range from 0 to 100, with pLDDT ≥ 90 indicating a very high confidence residue prediction, whereas residues with $90 > \text{pLDDT} \geq 70$ are classified as high confidence prediction. Residues with $70 > \text{pLDDT} \geq 50$ are considered to have a low confidence prediction, and residues with a pLDDT < 50 are considered to have a very low confidence prediction.

The subunits with the highest pLDDT values were selected for subsequent antigen construction. Specifically, the adjuvant sequence was genetically fused to the N-terminus of the chimeric protein, followed by the sequential linkage of the three subunits in the order of B-cell, CTL, and HTL epitopes. Intervening peptide linkers connecting these elements were designed using the RFdiffusion method [36] to generate 500 backbone scaffolds with compatible three-dimensional geometries. Subsequently, the full-length fusion protein sequence was optimized using ProteinMPNN (v. 48.020 model) [37] with 10 iterations performed per scaffold to enhance its folding potential and stability. The predicted tertiary structure of the final chimeric antigen was generated by AlphaFold3 via its online server (<https://alphafoldserver.com/>) [38].

2.3. Secondary structure prediction

Secondary structure prediction for the AI-designed chimeric antigen was performed based on its primary sequence. The sequence was submitted to the PSIPRED 4.0 server (<https://bioinf.cs.ucl.ac.uk/psipred/>) [39], which predicts structure from sequence, under default settings. Subsequently, to assess the differences in secondary structure resulting from the structural refinement process, the three-dimensional models (in PDB format) from both pre- and post-refinement were uploaded to the PDBsum Generate server (<https://www.ebi.ac.uk/thornton-srv/databases/pdbsum/Generate.html>) [40]. This server extracts secondary structure assignments from the atomic coordinates of a PDB file. Predictions were likewise conducted with the server's default parameters.

2.4. Antigenicity, allergenicity, solubility, and physicochemical properties

The antigenicity of the multi-epitope antigen was predicted using VaxiJen v2.0 (<https://www.ddg-pharmfac.net/vaxijen/VaxiJen/VaxiJen.html>) [41] and ANTIGENpro (<https://scratch.proteomics.ics.uci.edu/>) [42,43], with a threshold of > 0.5 for both. Solubility was evaluated using SoluProt (<https://loschmidt.chemi.muni.cz/soluprot/>) [44] and SOLpro (<https://scratch.proteomics.ics.uci.edu/>) [43,45], also applying a threshold of > 0.5 . For allergenicity assessment, AllerTOP 2.0 (<https://webs.iitd.edu.in/raghava/algpred2/batch.html>) [46] was used with a threshold of < 0.3 . The physicochemical properties, including the instability index (< 40) and isoelectric point, were analyzed using the ExPASy ProtParam tool (<https://web.expasy.org/protparam/>) [47].

2.5. Tertiary structure refinement and validation

The “rough” 3D model of the multi-epitope antigen obtained from AlphaFold3 was refined using ModRefiner with default parameters [48]. ModRefiner used C α traces to guide protein construction and refinement, and finally obtained a single refined model through a two-step atomic-level energy minimization process. The structure validation was conducted using PROCHECK (<https://www.ebi.ac.uk/thornton-srv/software/PROCHECK/>) [49], MolProbity (<https://molprobity.biochem.duke.edu/>) [50], and ProSA-web (<https://prosa.services.came.sbg.ac.at/prosa.php>) [51], respectively. For PROCHECK analysis, a high-quality model is typically defined as having over 90 % of residues in the most favored Ramachandran regions, while for MolProbity, the corresponding threshold is > 98 % in the favored regions.

2.6. Molecular docking

To evaluate the immunogenic potential of the designed antigen, HDock-based molecular docking simulations were conducted (<http://hdock.phys.hust.edu.cn/>) [52]. The refined vaccine construct served as the ligand, while Toll-like receptors TLR2 (PDB: 2Z7X) and TLR4 (PDB: 8WTA) were employed as receptors. For each receptor-ligand pair, five independent docking models were generated to systematically evaluate binding interactions. The optimal model was selected through a dual-criteria approach: (1) specificity of interaction, with a focus on models demonstrating binding at the interface between the adjuvant region of the chimeric antigen and the TLR; and (2) binding confidence, where the model with the highest confidence score among those that satisfied the specificity criterion was selected. This model was ultimately regarded as the most probable interaction model for subsequent analysis.

To analyze the detailed interactions between the vaccine candidate and immune receptors, the PDBsum server [40] was used with default parameters. Furthermore, to quantify the binding affinity of the vaccine with TLR2 and with TLR4, a comprehensive analysis was performed using PRODIGY (<https://rascar.science.uu.nl/prodigy/>) [53], with the temperature set to 37°C.

2.7. Molecular dynamics simulation

Molecular dynamics simulations of the chimeric antigen in complex with TLR2 or TLR4 were conducted using the GROMACS 2024.5 package [54], with the refined tertiary structure serving as the starting reference. Each complex system was solvated in a cubic box with periodic boundary conditions (PBC) and a minimum distance of 10 Å between the solute and the box edges. The TIP3P water model was used for solvation, with 109,341 and 124,729 water molecules added to the antigen/TLR2 and antigen/TLR4 systems, respectively. To neutralize the systems and maintain a physiological salt concentration (0.15 M NaCl), 318 Na⁺ and 350 Cl⁻ ions were introduced for the antigen/TLR2 complex, and 364 Na⁺ and 382 Cl⁻ ions for the antigen/TLR4 complex.

Energy values were recorded every 1.2 ps, and trajectory snapshots were saved every 100 ps. Both systems were equilibrated in the NPT ensemble at 300 K and 1 bar. Production simulations were carried out for 100 ns using the OPLS-AA force field. Following the simulations, the root mean square deviation (RMSD) of the protein complexes was evaluated over the entire 100-ns trajectories, while the root mean square fluctuation (RMSF) was calculated using the entire trajectory to ensure comprehensive sampling across all simulated timeframes.

2.8. Immune simulation

To assess the immune response profile and immunogenicity of the vaccine, *in silico* simulations were conducted using C-ImmSim (<https://150.146.2.1/C-IMMSIM/index.php>) [55]. The simulation parameters were set as follows: a simulation volume of 10 μ L and an antigen quantity of 1000 proteins per injection, based on the default settings recommended by the C-ImmSim server. The simulation consisted of three injections, administered four weeks apart at time steps 1, 84, and 168 (each step corresponding to 8 h in real life, with time step 1 representing the injection time, $t = 0$), to mirror standard three-dose immunization protocols used in experimental vaccine studies. The number of simulation steps was set to 1050, with a random seed of 12,345 for reproducibility. For HLA supertype selection, MHC-I alleles A*02:01, A*24:02, B*07:02, and B*44:02, and MHC-II alleles DRB1*01:01 and DRB1*03:01 were chosen based on the epitope profile of the designed chimeric antigen to ensure accurate simulation of its immunogenicity across diverse populations.

3. Results

3.1. Screening of highly effective epitopes for chimeric antigen construction

Numerous TB subunit vaccines have been developed, typically by combining various antigenic proteins [6–8]. Based on a comprehensive analysis of their immunogenicity and mechanisms of action, we selected eight key antigens for multi-epitope vaccine construction, including Ag85A, Ag85B, ESAT-6, CFP10, Hsp65, Mpt64, HspX, and Hsp70 (Table S1). These antigens represent major secretory proteins, heat shock molecular chaperones, and key virulence factors that are able to effectively induce strong T-cell and/or B-cell immune responses. Pre-clinical and clinical studies have demonstrated that they induce protective immunity against both active and latent TB [7,8]. Incorporating them into a single vaccine design could synergistically enhance immune recognition across different stages of Mtb infection.

To identify potent epitopes for multi-epitope vaccine construction, we performed a comparative epitope analysis [32] of these eight antigens, focusing on experimentally validated epitopes. This approach revealed a diverse repertoire of CTL (Fig. 1A), HTL (Fig. S1A), and linear B-cell epitopes (Fig. S1B). We prioritized epitopes with strong experimental support for further manual curation, selecting 21 CTL (Fig. 1A; Table S2), 21 HTL (Fig. S1A; Table S3), and 19 B-cell epitopes (Fig. S1B; Table S4) as high-efficacy candidates (indicated by arrows). For CTL epitopes, we assessed binding affinity to MHC class I molecules, MHC-I allele specificity, IFN- γ release activity, and cytotoxicity. For HTL epitopes, we evaluated their capacity to induce key cytokines, including IFN- γ , IL-4, IL-2, TNF- α , and IL-13. For B-cell epitopes, we examined binding affinity to BCRs and protective efficacy against pathogen challenges. Following this evaluation, we selected 17 highly potent epitopes, including 7 CTL epitopes, 5 HTL epitopes (including the universal PADRE epitope), and 5 B-cell epitopes (Fig. 1B). These epitopes, derived from eight distinct antigens (Fig. S2), target major MHC-I alleles (e.g., A*02, A*03, A*24, B*07) covering over 90 % of the global population [56]. Consequently, a vaccine incorporating these epitopes represents a potential universal solution for TB.

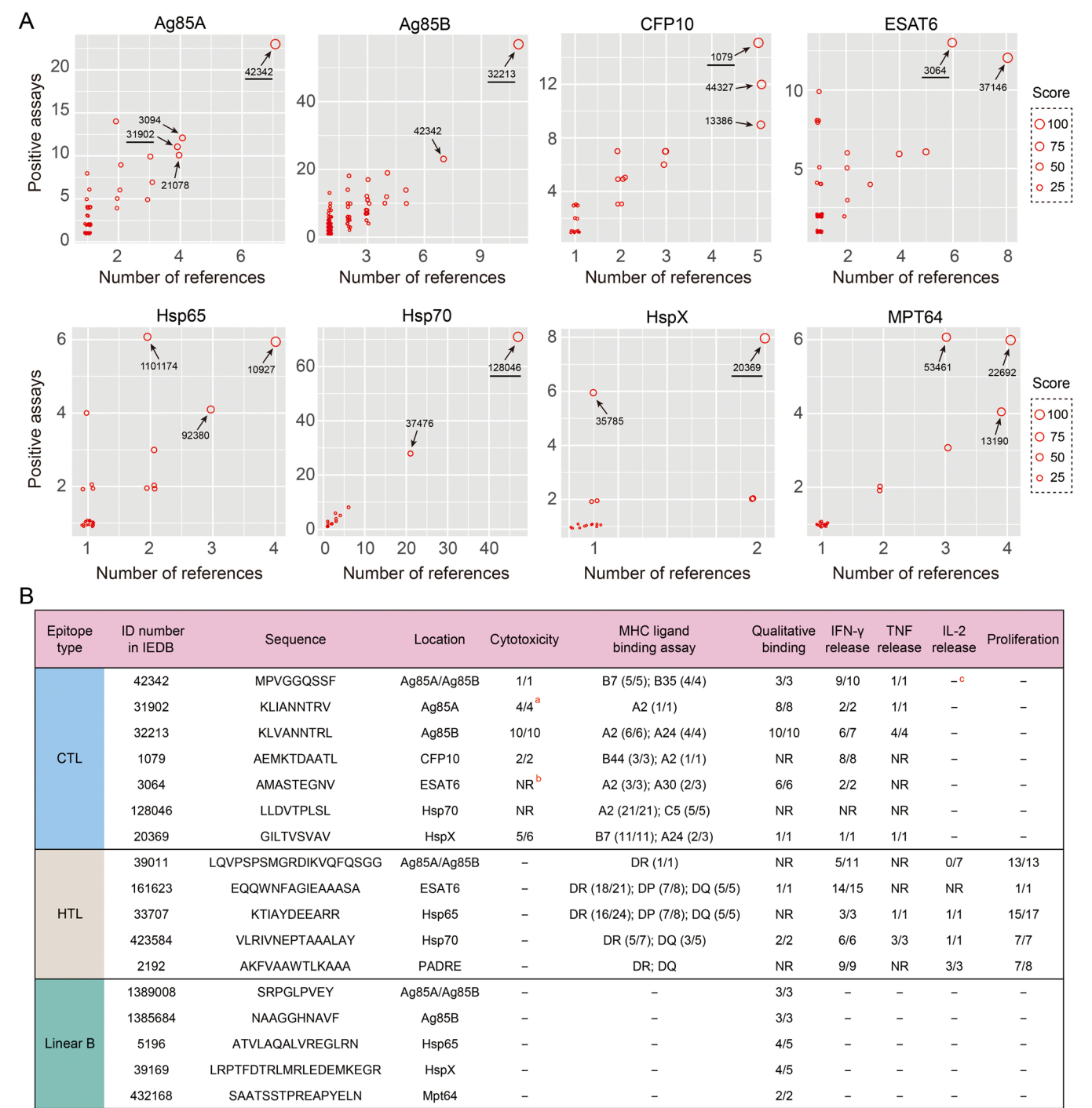


Fig. 1. Screening of potent epitopes for the construction of multi-epitope vaccines. (A) Characteristics of the experimentally validated cytotoxic T lymphocyte (CTL) epitopes identified from the eight major Mtb antigens (Ag85A, Ag85B, CFP10, ESAT-6, Hsp65, Hsp70, HspX, and Mpt64) using a comparative epitopomics approach. Each circle represents an epitope, and its size reflects its enrichment score. The epitopes marked with arrows represent initially screened candidates with potential high efficacy, with their corresponding accession numbers in the IEDB indicated. The underlined epitopes denote the potent epitopes selected for the construction of the multi-epitope vaccine. (B) Detailed information of the final selected epitopes. Symbols: a: "Ratio format (e.g., "4/4") indicates "number of positive assays/total number of assays"; b: "NR" denotes "Not Recorded"; c: "-" indicates not applicable.

3.2. Rational design of multi-epitope antigens via AI-driven protein design

When designing multi-epitope antigens, the selection of adjuvant, epitope flanking sequences, and epitope arrangement are key factors affecting the immunogenicity and stability of antigen. Human β -defensin-2 (hBD-2) and hBD-3 were selected as adjuvants for their ability to interact with immune receptors such as TLRs and CCR6, thereby effectively enhancing innate and adaptive immunity [57–59]. To enhance the

efficiency of epitope processing and presentation, the "AAY" linker was used to connect CTL epitopes because it promotes the generation of appropriate processing sites for CTL epitopes [29]. Similarly, the "GPGPG" linker was selected to connect HTL epitopes because it supports epitope processing and presentation in endosomes. In addition, the two-residue "KK" linker was used to connect various B-cell epitopes because it prevents the generation of new epitopes in the linker region. Regarding the construction of chimeric antigen, we initially used

random ligation to directly link various epitopes. However, the resulting chimeric protein had poor structural integrity and low stability (Fig. 2). This low quality likely arises because epitopes typically require specific sequence contexts or local structural environments for proper folding; direct fusion of epitopes with incompatible physicochemical properties may disrupt these requirements. Therefore, in addition to linkers that facilitate epitope processing, incorporating auxiliary sequences may be essential to maintain structural stability. Capitalizing on recent advances in AI-driven protein design [34,60,61], we employed a deep learning-based approach. Specifically, we grouped the 17 epitopes into three different functional modules: a CTL epitope subunit, an HTL epitope subunit, and a B-cell epitope subunit. Each module was individually optimized by the ProteinGenerator algorithm to determine the optimal epitope order and flanking sequences. As a result, AI-optimized subunits exhibited dramatically enhanced structural integrity and quality compared to subunits generated by random ligation (Fig. 2B–D). To assemble the full-length construct, we fused the three subunits to an adjuvant protein and designed the linking peptides using RFDiffusion with side-chain conformations refined by ProteinMPNN. This process yielded a final recombinant protein, which we named MtbEpi-17, with superior structural integrity and rationality compared to the initial random ligation construct (Fig. 2E–F). The final configuration of MtbEpi-17 is summarized in Fig. 3S.

Next, we compared the stability, structural reliability, and rationality of the AI-designed chimeric protein (MtbEpi-17) with a randomly assembled chimeric protein. In terms of protein stability, we analyzed the instability index of 100 randomly assembled chimeric proteins and found that their instability indices were all higher than 40, classifying them as unstable proteins (Fig. 2G). In contrast, the instability index of protein-engineered MtbEpi-17 was less than 40, indicating higher stability. To evaluate the quality and rationality of the protein structure, we used the PROCHECK, MolProbity, and ProSA-web tools. PROCHECK is mainly used to assess the geometric quality of protein structures, especially the stereochemical properties of the backbone. The analysis showed that 95.1 % of the amino acid residues in MtbEpi-17 were located in the most favored regions (A, B, L) of the Ramachandran plot, while only 81.8 % of the residues in the randomly assembled chimeric proteins were located in these regions (Fig. 2H). In addition, 3.6 % of the residues in the randomly generated proteins were located in disallowed region, whereas only 0.3 % of the residues in MtbEpi-17 were in these regions (Fig. 2H). Similar results were obtained from the analysis in MolProbity (Fig. 2I), a comprehensive tool that provides detailed geometry checks, stereochemistry assessments, and hydrogen bond evaluations. Specifically, 83.5 % and 94.4 % of the residues in the randomly assembled chimeric protein were located in the favored and allowed regions, respectively. In contrast, these proportions in MtbEpi-17 increased to 94 % and 99.1 %, respectively (Fig. 2I). To further assess the accuracy of the predicted protein structure, we also carried out ProSA-web analysis. This tool primarily evaluates the overall energy state of the protein structure and provides a Z-score that reflects the likelihood of the structure being correct. The results from ProSA-web were consistent with those from PROCHECK. Specifically, the Z-score of the randomly assembled protein was -4.18 , which fell outside the range defined by the reference set of experimentally determined protein structures in ProSA-web [51]. In contrast, the Z-score of MtbEpi-17 was -10.32 , falling within the typical Z-score range of these experimentally determined proteins (Fig. 3S4). Taken together, these results indicate that MtbEpi-17 designed using AI-driven protein engineering is significantly superior to randomly generated chimeric proteins in terms of structural integrity and stability, suggesting that it is a potent candidate antigen for Mtb.

3.3. Refinement and validation of the tertiary structure of MtbEpi-17

To generate a more accurate atomic model, we refined the initial tertiary structure of MtbEpi-17 using ModRefiner [48], which builds

full-atom models from C α traces (Fig. 3). This refinement introduced noticeable local improvements to both the secondary (Fig. 3S5) and tertiary (Fig. 3A–C) structures. Validating the model is a critical step for identifying potential errors in predicted structures. We therefore employed three validation tools: PROCHECK, MolProbity, and ProSA-web. PROCHECK results showed an increase in residues within the most favored Ramachandran region from 95.1 % to 96.3 %, with a corresponding decrease in the additionally allowed region from 4.2 % to 3.1 % (Figs. 3D, 3G, and 2H). Similarly, MolProbity analysis indicated a substantial rise in residues in the favored region from 94 % to 98.1 % (Figs. 3E, 3G, and 2I). Furthermore, the ProSA-web Z-score improved from -10.32 to -11.37 post-refinement, which is closer to the range typical for experimentally determined structures (Fig. 3F, 3G, and 3S4B). The above results collectively indicate that although the initial prediction was already of high quality, the refinement process yielded a more robust and reliable structural model, appropriate for subsequent functional analysis.

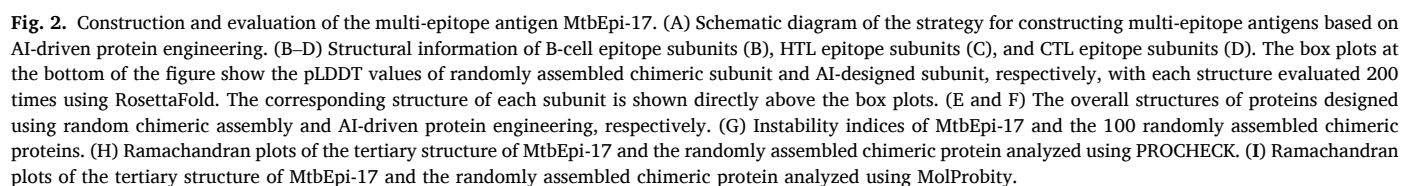
3.4. Molecular docking of MtbEpi-17 with immune cell receptors

The efficacy of a subunit vaccine hinges on its ability to be recognized by immune receptors to initiate a protective response. Given the pivotal role of Toll-like receptors (TLRs) in launching innate immunity and orchestrating adaptive responses, we sought to determine if our multi-epitope antigen, MtbEpi-17, could directly interact with TLR2 and TLR4—receptors known to recognize viral proteins and trigger essential inflammatory signaling pathways [62,63]. Ligand engagement of these receptors on dendritic cells activates them, leading to the stimulation of naïve T cells and the secretion of critical immune factors like IFN- γ and IL-4 from CD4 $^{+}$ and CD8 $^{+}$ T cells. Our molecular docking analysis using the HDock server predicted robust and stable binding between MtbEpi-17 and both receptors (Fig. 4A). The top computational model positioned MtbEpi-17 into the binding site of TLR2 with high confidence (score: 0.9731) and a strong docking score of -329.36 (Fig. 4A and C). Similarly, docking to TLR4 yielded a reliable complex (confidence score: 0.936) with a favorable energy score of -284.17 (Fig. 4B and C).

A detailed dissection of the binding interfaces provided a structural basis for this high affinity. The MtbEpi-17/TLR2 complex was notably stabilized by an extensive network of interactions, including 8 hydrogen bonds and 3 salt bridges across 33 and 25 residues, respectively (Fig. 4C–E). The interaction with TLR4, while involving fewer hydrogen bonds (3), was also extensive, engaging 25 residues from each protein. Moreover, we calculated the binding affinity using PRODIGY [53]. The results confirmed a high affinity for both complexes. The dissociation constant (Kd) for MtbEpi-17/TLR2 was calculated to be 2.5×10^{-10} M ($\Delta G = -13.6$ kcal/mol), while the binding to TLR4 was marginally stronger (Kd = 1.7×10^{-10} M; $\Delta G = -13.9$ kcal/mol) (Fig. 4C). This potent, high-affinity interaction with two key immunostimulatory receptors provides a compelling mechanistic hypothesis for how MtbEpi-17 likely triggers the robust immune responses observed in our functional studies.

3.5. Molecular dynamics simulation of MtbEpi-17/TLR2/TLR4 complexes

To evaluate the stability of the complexes, molecular dynamics (MD) simulations were performed (Fig. 5). Trajectory analysis was carried out by calculating the root mean square deviation (RMSD) and root mean square fluctuation (RMSF), with all trajectories aligned to the initial reference frame for comparative assessment (Fig. 5A and D). In the case of the MtbEpi-17/TLR2 complex, the RMSD values increased rapidly during the initial 10 ns, followed by a slow increase between 10 and 30 ns (Fig. 5B). After 30 ns, the system entered a stable equilibrium, with the RMSD exhibiting minor fluctuations around a consistent plateau. A similar trend was observed for the MtbEpi-17/TLR4 complex. The RMSD initially increased during the first 20 ns, followed by an equilibration phase where the value stabilized and fluctuated within a



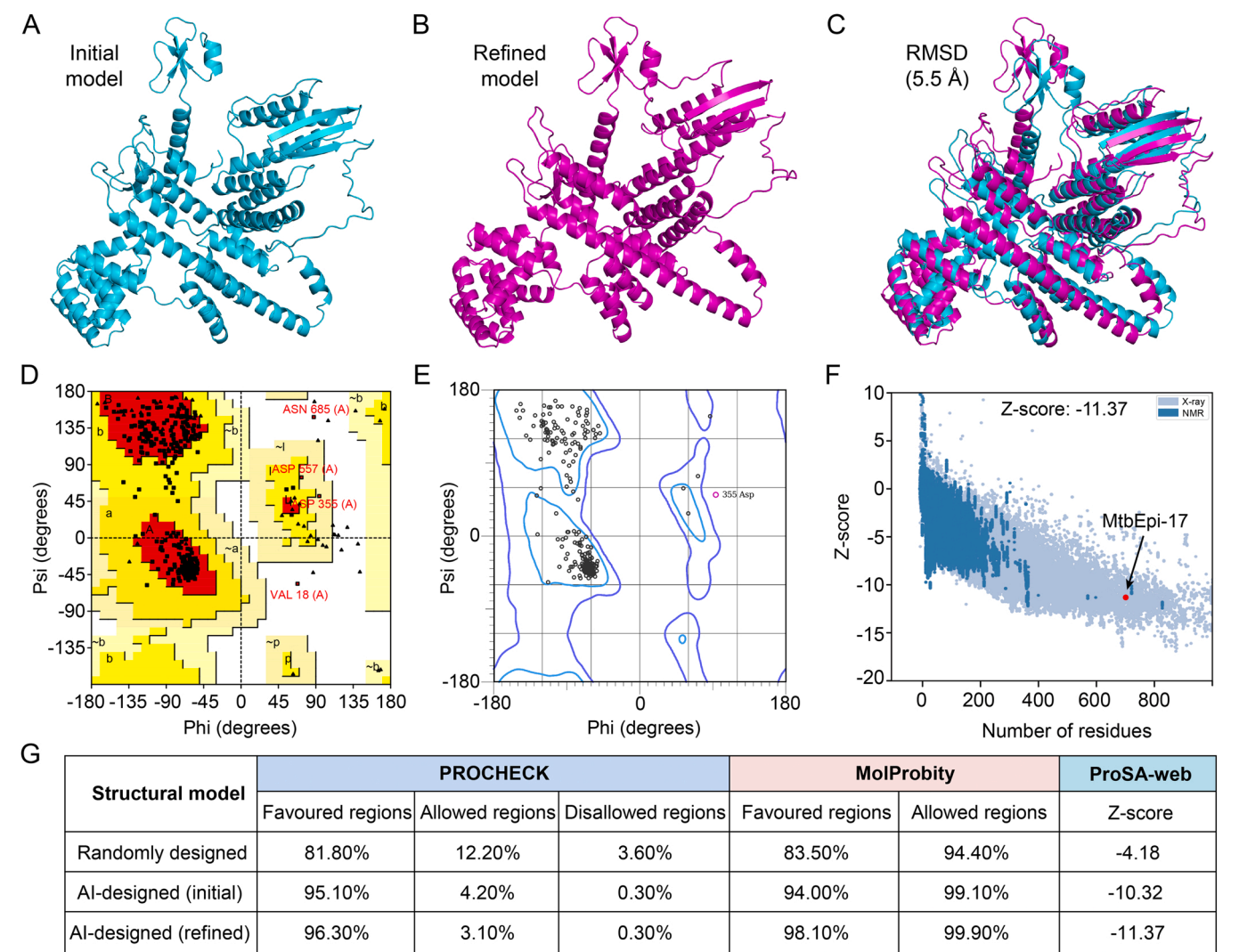


Fig. 3. Refinement and validation of the tertiary structure of MtbEpi-17. (A) The initial tertiary structure of MtbEpi-17 predicted by the AlphaFold3 server. (B) The final tertiary structure of MtbEpi-17 refined using ModRefiner server. (C) Alignment of the initial and refined structures of MtbEpi-17. (D) Ramachandran plot of the refined tertiary structure of MtbEpi-17 analyzed using PROCHECK. (E) Ramachandran plot of the refined tertiary structures of MtbEpi-17 analyzed using MolProbity. (F) Validation of the refined structures of MtbEpi-17 using ProSA-web server. Each blue dot represents the Z-score of a reference protein structure from the experimentally determined set in ProSA-web. (G) Summary of key structural metrics for randomly designed, AI-designed (initial), and AI-designed (refined) protein models, as assessed by PROCHECK, MolProbity, and ProSA-web.

narrow range for the remainder of the simulation (Fig. 5E).

RMSF analysis revealed a stable pattern for TLR-interacting residues, with key interface residues maintaining low fluctuation values (0.1–0.4 nm) throughout the 100 ns simulation (Fig. 5C and F). In the TLR2 complex, hydrogen bond-forming residues (e.g., Arg94, Thr337, Tyr373, and Ala463) and van der Waals interaction residues (e.g., Val64, Arg68, Gly423, and Leu425) exhibited particularly stable behavior (Fig. 5C), while the TLR4 complex showed similar stability, with RMSF values for key residues (e.g., Val366, Arg89, Gly451, and Ala452) consistently remaining below 0.4 nm (Fig. 5F). Notably, the N-terminal adjuvant region (residues 50–100) of both complexes exhibited low fluctuation (RMSF < 0.4 nm), confirming its structural integrity during binding. Importantly, no interface residues in either complex exceeded 0.5 nm RMSF, strongly demonstrating the sustained structural stability of the MtbEpi-17’s receptor-binding interface. These results are consistent with previous observations and collectively suggest that MtbEpi-17 exhibits a strong affinity for both TLR2 and TLR4, thereby enabling the induction of a robust and stable immune response.

3.6. MtbEpi-17 may elicit strong innate and adaptive immune responses

To evaluate the immunogenicity of MtbEpi-17, we performed an *in silico* immune simulation using the C-ImmSim server. Following immunization, the immune response was initiated by the activation of dendritic cells (DCs), which, as antigen-presenting cells (APCs), are pivotal for priming T helper (Th) cells. As shown in Fig. 6A, antigen presentation by DCs increased rapidly after the first dose. Th cell activation was observed shortly thereafter, with both active and memory Th cell populations expanding (Fig. 6B and C). These responses were significantly boosted by the second immunization and marginally enhanced by the third. Since Th cells are essential for B cell activation and antibody production, we also monitored B cell dynamics. Active B cell counts rose substantially after each immunization and continued to accumulate throughout the regimen (Fig. 6D). Consistent with Th and B cell activation, antibody titers particularly IgM and IgG1 increased markedly following the second and third immunizations (Fig. 6E). Under IFN-γ stimulation, Th cells differentiate into Th1 cells, which secrete cytokines such as IFN-γ and IL-2 to activate cytotoxic T (Tc) cells and macrophages, thereby promoting cell-mediated immunity. Both

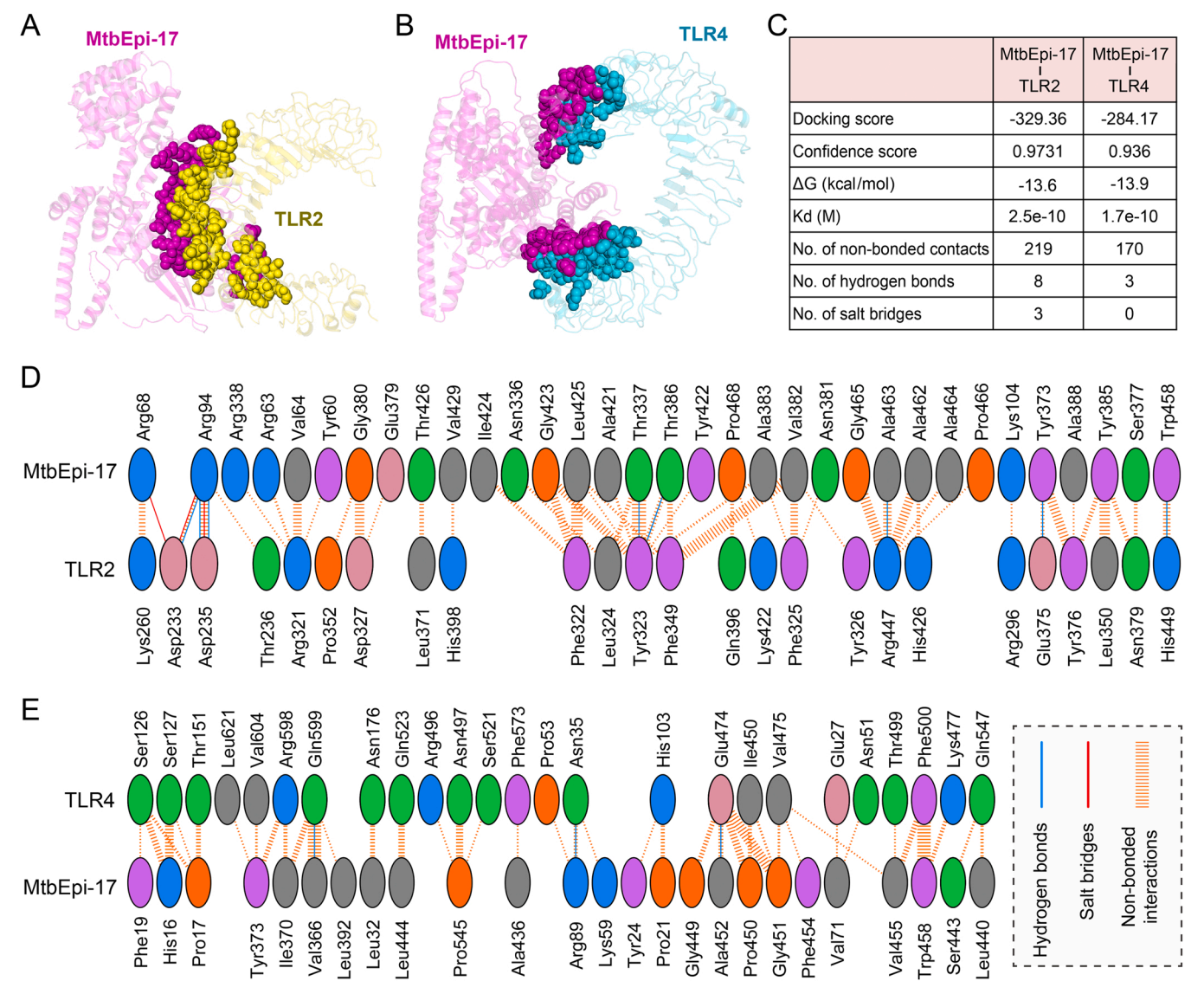


Fig. 4. Molecular docking of MtbEpi-17 with immune cell receptors. (A and B) Potential interactions of MtbEpi-17 with TLR2 (A) or TLR4 (B). The overall structure of MtbEpi-17, TLR2, and TLR4 are shown in carton with 80 % transparency, and the interface regions are shown in spheres. (C) Interaction parameters between MtbEpi-17 and TLR2/TLR4. (D and E) Specific interacting residues and interaction forces between MtbEpi-17 and TLR2/TLR4. The colors of interacting residues represent the nature of amino acids (positive: blue; negative: pink; neutral: green; aliphatic: grey; aromatic: purple; Pro and Gly: orange; Cys: yellow).

IFN- γ and IL-2 secretion increased significantly after each immunization round (Fig. 6F). Accordingly, Tc cells (Fig. 6G) and macrophages (Fig. 6H) were effectively stimulated during vaccination. Active Tc cells proliferated rapidly after the first dose, maintained high levels after boosting, and gradually declined thereafter. Resting Tc cells displayed a converse trend, rising initially before gradually decreasing. Memory and anergic Tc cell levels remained relatively stable throughout (Fig. 6G). Meanwhile, macrophages became activated early in the immune response and sustained high activity levels during the immunization period. After the third dose, activated macrophages remained functional for several days before returning to rest, potentially mitigating excessive immunopathology and inflammation (Fig. 6H). We also assessed the effect of MtbEpi-17 on natural killer (NK) cells, key mediators of innate immunity. The NK cell population remained stable during immunization (Fig. 6I), suggesting innate immunity was maintained.

To further validate that the observed immune responses were specifically elicited by MtbEpi-17 and not due to arbitrary sequence features, we conducted a negative control simulation using a randomly generated sequence. As shown in Fig. S6, this random sequence failed to trigger significant humoral or cellular immune responses, in contrast to

the robust activation induced by MtbEpi-17. This result confirms that the immunogenicity of MtbEpi-17 is attributable to its rationally selected epitope composition. Collectively, these simulations suggest that MtbEpi-17 can potentially induce both innate and adaptive immune responses, supporting its promise as a vaccine candidate.

4. Discussion

The continuous evolution of pathogens and the emergence of novel infectious diseases increasingly challenge the efficacy of traditional antigen-based vaccines, which are often limited in their ability to address these evolving threats to human health [64,65]. High-efficacy vaccines are therefore crucial not only for mitigating the impact of outbreaks but also for minimizing the risks of drug resistance and immune escape. Multi-epitope vaccines offer a promising strategy for developing potent, broad-spectrum immunogens by overcoming the inherent limitations of single antigens through the incorporation of multiple protective epitopes [56,66]. In this study, we selected eight of the most commonly utilized TB vaccine antigens. Leveraging a comparative epitopomics approach, we successfully identified 17 highly

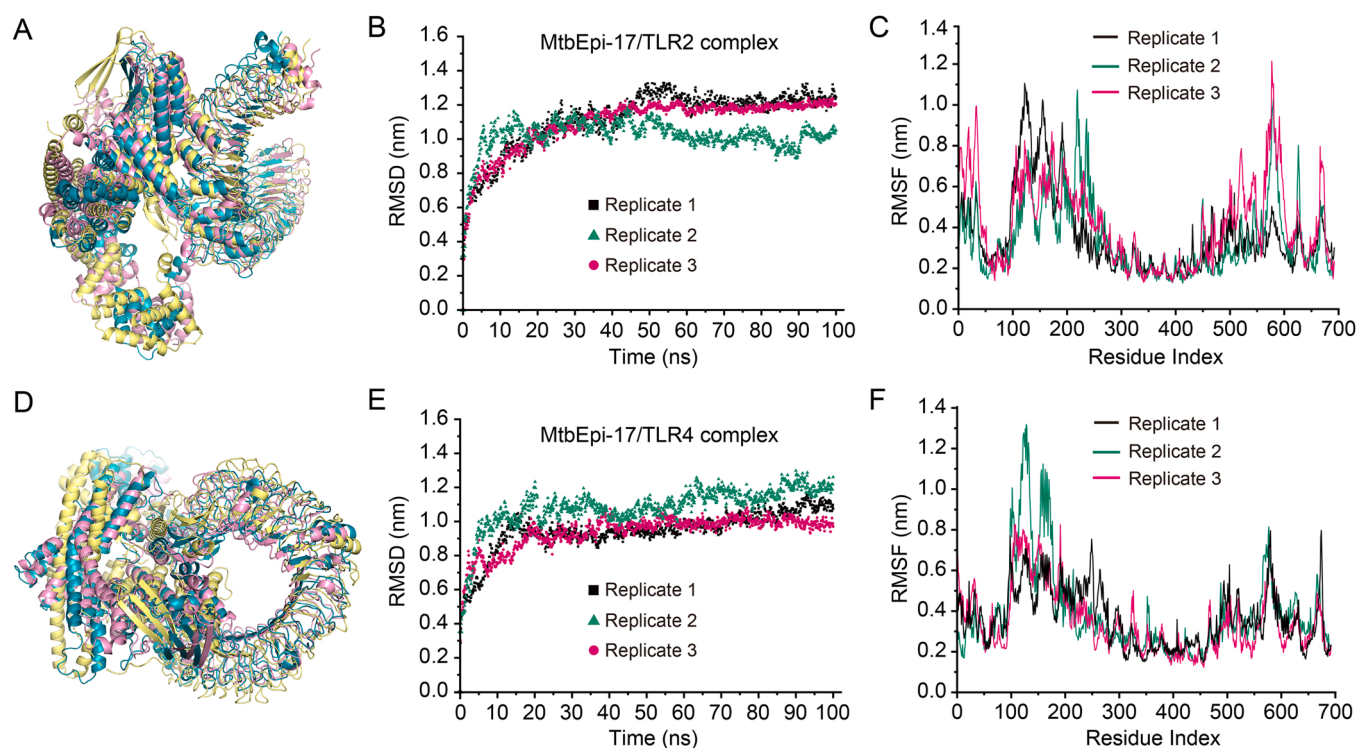


Fig. 5. Molecular dynamics simulations of the MtbEpi-17/TLR2 and MtbEpi-17/TLR4 complexes. (A and D) Superimposition structures of the MtbEpi-17/TLR2 (A) or MtbEpi-17/TLR4 (D) complex obtained from MD simulation trajectories. Three conformations of the superimposed states were extracted from the dynamic trajectories of 0.1, 50, and 100 ns, respectively. (B and E) RMSD plots of the MtbEpi-17/TLR2 (B) and MtbEpi-17/TLR4 (E) complexes, reflecting the overall stability of the bound systems. (C and F) RMSF plots of MtbEpi-17 within the MtbEpi-17/TLR2 (C) and MtbEpi-17/TLR4 (F) complexes, showing residue-specific flexibility. Three independent replicates are displayed for each analysis.

potent epitopes from these antigens (Fig. 1B). These epitopes collectively provide coverage for over 90 % of prevalent global MHC-I molecules, theoretically enabling them to target both active and latent TB infections. We then employed an AI-driven protein design platform to iteratively optimize the order and flanking sequences of these epitopes, ultimately engineering a structurally stable and well-folded chimeric antigen we named MtbEpi-17 (Fig. 2). Subsequent computational analyses further indicated that MtbEpi-17 interacts with key immune receptors TLR2 and TLR4, suggesting its potential to efficiently stimulate both innate and adaptive immune responses (Figs. 4–6).

The selection of highly potent epitopes is paramount in multi-epitope vaccine design, as it directly dictates the vaccine's immunogenicity. While many current studies rely primarily on bioinformatic predictions for epitope identification [22,24–27], predicted epitopes may have poor performance compared to those validated experimentally. The incorporation of experimentally verified epitopes can thus significantly enhance vaccine efficacy. A notable aspect of our work is the application of our recently developed comparative epitopomics platform, a robust method for the efficient screening of such validated epitopes [32]. Here, for the first time in the context of TB, we systematically integrated these experimentally confirmed epitopes into the design of a chimeric Mtb antigen. The resulting candidate antigen demonstrated strong immunogenicity in our models (Fig. 6), establishing a solid foundation for subsequent experimental validation. While our computational assessments suggest that MtbEpi-17 has strong immunogenic potential, we acknowledge that global HLA diversity could pose challenges for clinical translation. Although our selected epitopes encompass coverage for more than 90 % of global MHC-I molecules (including prevalent alleles such as A*02, A*03, A*24, and B*07), regional variations in HLA distribution may compromise vaccine efficacy in certain populations [67–69]. To address this, future clinical development may necessitate population-specific epitope optimization to maximize vaccine

responsiveness across diverse genetic backgrounds.

In addition to epitope selection, the linkage of different epitopes is also crucial in the construction of multi-epitope vaccines because it directly affects the stability of the vaccine construct. Currently, due to the lack of established guidelines of chimeric antigen design, most multi-epitope vaccine studies randomly connect epitopes [18,22,70–74], but ignoring the optimization of epitope order and flanking sequences. As a result, the resulting antigens typically have poor structural integrity and low stability. To address this challenge, we applied AI-driven protein design in the construction of the chimeric antigens. Computational analysis showed that protein-engineered antigen exhibited significantly enhanced conformational integrity and stability compared with randomly assembled antigens (Fig. 2). To the best of our knowledge, this represents the first application of AI-driven protein design technology specifically for TB multi-epitope vaccine development. Our strategy integrates comparative epitopomics with structural optimization algorithms, offering a novel paradigm for generating immunogenic chimeric antigens with enhanced stability.

It is important to note that, although AI-driven design of MtbEpi-17 shows great potential, some inherent limitations of current computational tools must be considered. First, the training data of algorithms such as ProteinGenerator, ProteinMPNN, and AlphaFold3 are biased toward natural protein sequences and folds [34,37,75]. This may limit their accuracy when applied to non-native chimeric constructs (such as our multi-epitope vaccine), particularly in two respects: (1) steric clashes or thermodynamic instability at the artificial epitope–linker junction may be underestimated, and (2) confidence metrics for de novo designed regions may be unreliable. To mitigate these issues, we implemented a computational validation pipeline that included multi-scale refinement (ModRefiner) and rigorous quality evaluation (ProSA-web, MolProbity, PROCHECK), which together produced improved structural parameters (Figs. 3, S4). Second, although AI-based

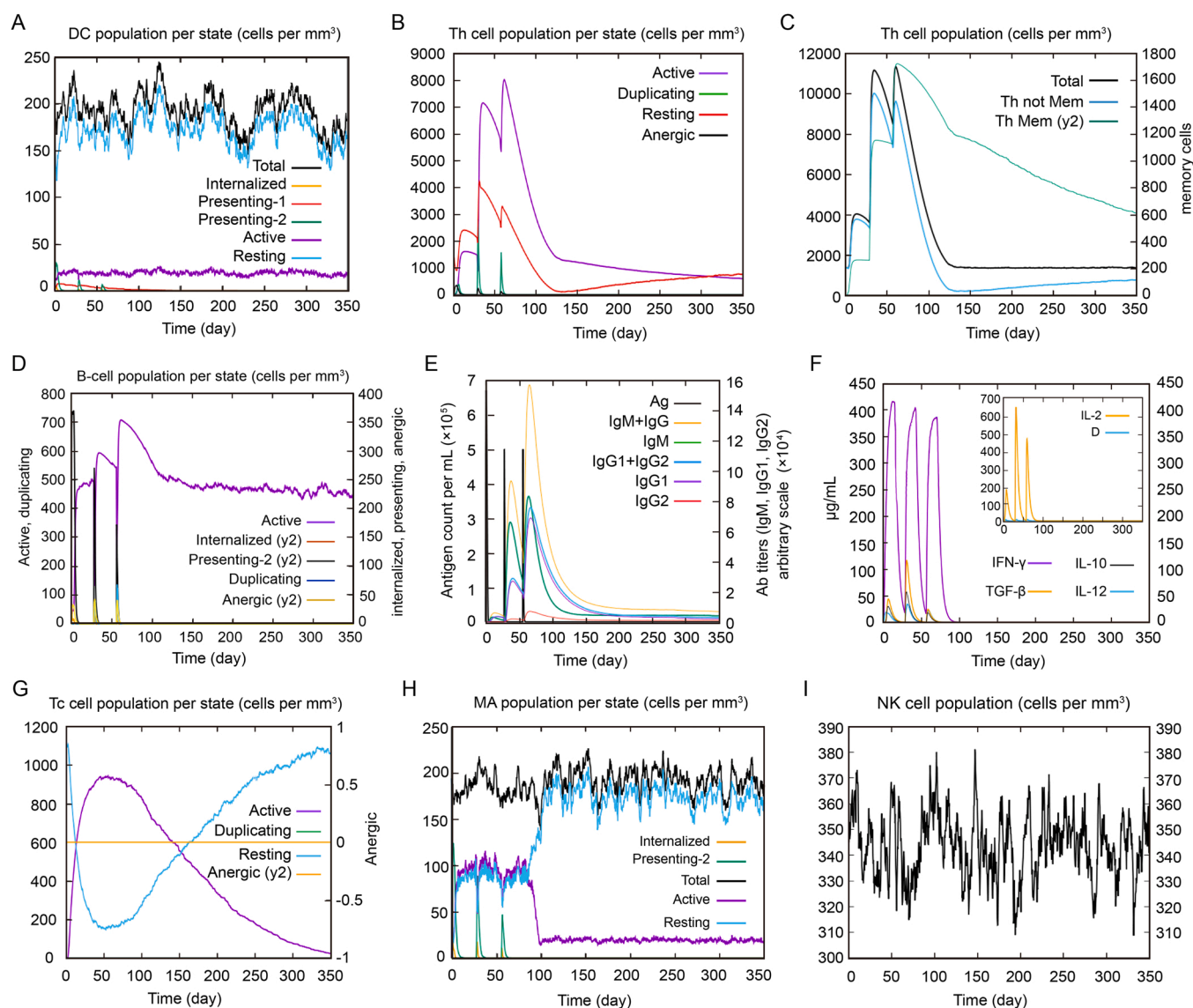


Fig. 6. Immunogenicity profile of MtbEpi-17. Three injections were administered on days 0, 28, and 56 (spaced four weeks apart). (A) Dendritic cell counts by functional state. (B) CD4⁺ T-helper (Th) cell counts in various activation states. (C) Total Th cell population dynamics. (D) B lymphocyte counts across different states. (E) Immunoglobulin production and temporal variation following MtbEpi-17 injection. (F) Cytokine and interleukin concentration kinetics; the inset “D” indicates the timing of danger signal introduction. (G) CD8⁺ T-cytotoxic (Tc) cell counts per state. (H) Macrophage counts by functional state. (I) Natural killer (NK) cell population dynamics. Key definitions: Internalized: Cells that have internalized and are processing antigen; Resting: Resting cells that are not participating in immune activity; Duplicating: Cells undergoing clonal expansion after activation; Anergic: cells that are functionally unresponsive after exposure to antigen; Presenting-1 / Presenting-2: cells that present processed antigen via MHC class I or class II molecules, respectively; Th Mem: Memory helper T cells, mediating rapid recall responses; (y2): Data are plotted on a secondary y-axis (right side) due to scaling differences.

methods can effectively explore sequence space, they do not fully capture the complex biophysics of protein folding under physiological conditions [76,77]. This highlights the need to complement *in silico* predictions with binding simulations (Figs. 4 and 5) and immune profiling (Fig. 6). Critically, these computational results, while encouraging, must ultimately be validated through wet-lab characterization of MtbEpi-17’s structural stability, immunogenicity, and protective efficacy in animal models. Such experimental validation will not only test the real-world performance of our designs but also provide crucial feedback for improving next-generation AI tools for vaccinology.

To evaluate the potential immunostimulatory capacity of the vaccine, we used HDock to carry out molecular docking studies targeting vaccine-TLR interactions (Fig. 4). To improve the reliability of docking predictions, cross-validation using multiple software tools or complementary analysis methods is often recommended. In this study, the confidence of the molecular docking results was improved by the

following approaches: In addition to selecting the top-ranked conformations based on high confidence scores (0.9731 for TLR2 and 0.936 for TLR4), we also prioritized models that strongly interacted with human β -defensin-2/3 adjuvant, which is known to interact with TLRs [57–59]. Moreover, all docked complexes were subjected to molecular dynamics simulations up to 100-ns (Fig. 5), followed by quantitative evaluation of the complex stability and residue-specific interactions at the binding interface. Overall, these analyses provide strong support for the docking results.

While our computational analyses suggest that MtbEpi-17 exhibits promising structural stability and immunogenicity, experimental validation in a wet-lab setting is essential to confirm these predictions. To assess structural integrity, codon-optimized MtbEpi-17 constructs can be expressed and purified using *Escherichia coli* strain BL21(DE3). Circular dichroism (CD) spectroscopy (190–260 nm) can be used to characterize its secondary structure composition (e.g., α -helix/ β -sheet ratio) [78,79],

while differential scanning calorimetry (DSC) can provide insights into its thermal stability through melting temperature (T_m) measurements [80,81]. To assess immunogenicity, we recommend the use of HLA-transgenic mouse models, which are particularly important because the incorporated epitopes are of human origin [82,83]. Humoral immune responses can be assessed by enzyme-linked immunosorbent assay (ELISA) to determine antigen-specific antibody titers and subclass distribution [84,85]. Cellular immunity, including epitope-specific T-cell activation and cytokine production (e.g., IFN- γ , IL-2, and IL-4), can be quantified by flow cytometry and enzyme-linked immunospot (ELISpot) assays [86–88]. To further evaluate vaccine efficacy, MtbEpi-17-immunized mice could be challenged with a virulent Mtb strain such as H37Rv via an aerosol infection model [89]. Protection can then be determined by quantifying bacterial burden and examining lung histopathology in these animals [90].

Notably, MtbEpi-17 exhibits synergistic potential with the licensed BCG vaccine through complementary immunological mechanisms that may substantially enhance overall immunoprotection. The BCG vaccine, which lacks the RD1 genomic region encoding key antigens such as ESAT-6 and CFP-10, does not effectively induce CD8⁺ T cell responses [91,92]. In contrast, MtbEpi-17 incorporates multiple epitopes derived from these antigens, enabling the activation of T cell clones that are not stimulated by BCG vaccination and thereby enhancing cytotoxic T cell activity. Furthermore, while BCG primarily elicits Th1-type immune responses (dependent on CD4⁺ T cells and IFN- γ) [5], our vaccine contributes complementary CD8⁺ T cell activation, resulting in more comprehensive immune protection. This complementary profile supports the potential for prime-boost strategies, in which BCG establishes a foundational Th1 immunity, followed by MtbEpi-17 boosting to amplify CD8⁺ T cell memory responses. Additionally, the modular design of our vaccine allows for dynamic epitope adjustments, which may help overcome BCG's limitations related to HLA diversity and variable protective efficacy [4,5]. It should be noted, however, that these theoretical advantages require empirical validation through preclinical co-administration studies to assess potential interference risks and determine optimal immunization protocols.

5. Conclusion

Through comparative epitopomics, we identified 17 experimentally validated potent epitopes from eight major Mtb antigens. Iterative optimization of epitope order and flanking sequences using AI-driven protein design yielded a structurally stable chimeric antigen, termed MtbEpi-17. *In silico* analyses indicate that MtbEpi-17 exhibits high structural stability and antigenicity, and is predicted to effectively stimulate both innate and adaptive immune responses. Together, these findings establish a robust framework for multi-epitope vaccine design by integrating comparative epitopomics with AI-driven protein engineering. While these computational results are promising, the immunogenicity and protective efficacy of MtbEpi-17 against Mtb infection remain to be validated through subsequent wet-lab studies and clinical trials.

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CRediT authorship contribution statement

Xinfeng Li: Writing – original draft, Software, Conceptualization. **Xinyu Tao:** Validation, Software, Investigation. **Mingyue Zhong:** Visualization, Software. **Yiyao Wang:** Validation, Software. **Heng Xue:** Visualization, Investigation. **Binda T. Andongma:** Validation, Formal analysis. **Shan-Ho Chou:** Writing – review & editing, Formal analysis. **Hongping Wei:** Writing – review & editing, Formal analysis. **Jin He:**

Writing – review & editing, Supervision, Conceptualization. **Hang Yang:** Writing – review & editing, Supervision, Conceptualization.

Declaration of Competing Interest

The authors declare no competing interests.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.csbj.2025.09.015.

Data availability statement

All data generated or analyzed during this study are included in the manuscript and its supplementary files. Computational scripts are available from the corresponding author upon reasonable request.

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